TOPIC 1 · WHAT IOT AND OT ARE

5 minutes · smart devices at home, school and shop · OT in the community · the router as the front door

What is it?

IoT devices are ordinary objects with an internet connection: cameras, TVs, speakers, watches, plugs, doorbells, baby monitors. **OT** systems control physical processes: a society's water pump, a rooftop solar inverter, a lift, a fire alarm. Both are computers, but they are rarely treated as computers — so they are rarely updated or secured.



Figure 9.3 · A typical connected home

Setting	Common connected devices
Home	Router, smart TV, camera, smart speaker, doorbell, children's smartwatch, fitness band
School	Classroom displays, CCTV, biometric attendance, Wi-Fi access points, lab computers
Shop or clinic	CCTV, billing PC, POS terminal, UPI sound-box, medical devices, router
Housing society or village	Water-pump controller, gate access, intercom, solar inverter, lifts

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TOPIC 2 · WHY SMART DEVICES GET HACKED

5 minutes · default passwords · firmware · exposure · data collection · botnets · physical harm

What is it?

Most smart-device attacks need no special skill. Lists of default usernames and passwords are public, and automated programs scan the internet for devices that still use them. In 2016, the **Mirai** botnet took over hundreds of thousands of cameras and video recorders with default passwords and used them to knock major websites offline.

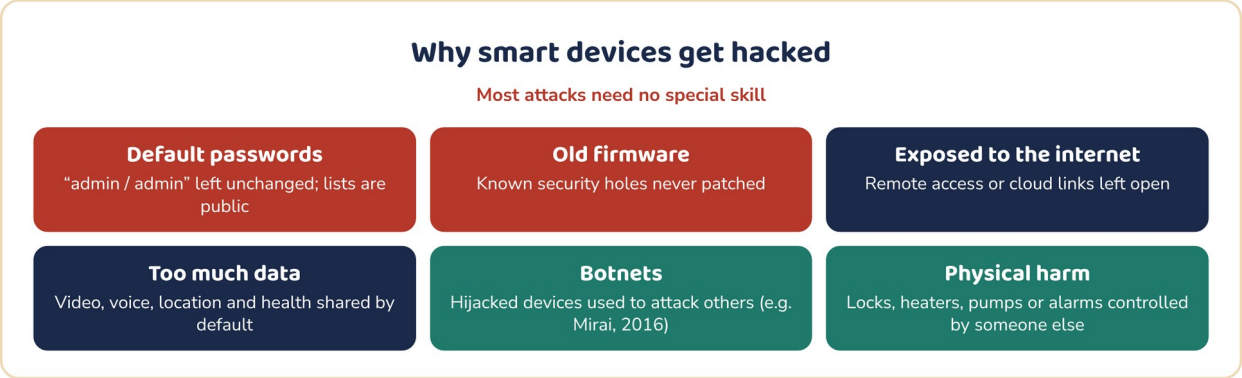


Figure 9.4 · Why smart devices get hacked

Remember

If a device came with a password printed on a sticker, change it the day you install it.

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TOPIC 3 · SECURING THE HOME OR SHOP ROUTER

5 minutes · admin password · Wi-Fi security · firmware · guest network · WPS, remote management and UPnP

What is it?

The **router** is the front door of every device behind it. Two passwords matter: the **Wi-Fi password** that devices use to join, and the **admin password** that opens the router's settings page. Many routers supplied by internet providers still use a default admin password.

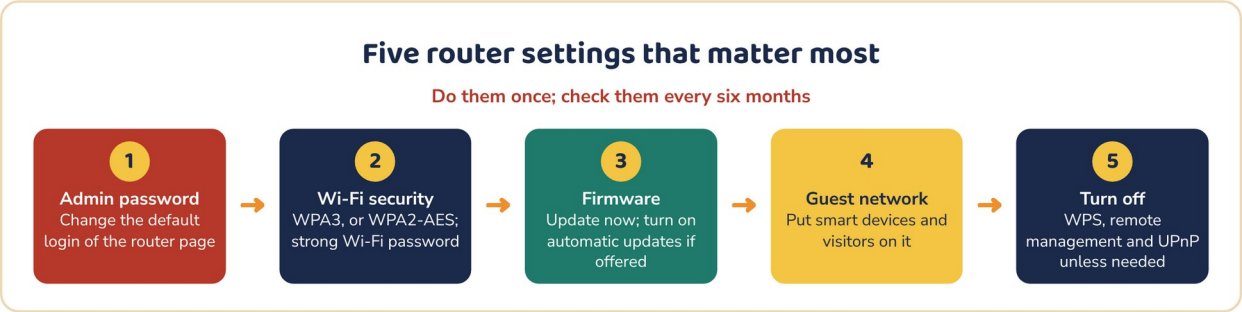


Figure 9.5 · Five router settings that matter most

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Setting	Where to find it (names vary by brand)	Good value
Admin password	Router page (often 192.168.0.1 or 192.168.1.1) → System / Administration	Unique passphrase, different from Wi-Fi password
Wi-Fi security	Wireless → Security	WPA3 or WPA2-AES (never WEP or “open”)
Firmware	System → Firmware upgrade	Latest version; automatic updates if offered
Guest network	Wireless → Guest network	On, for smart devices and visitors
WPS / remote management / UPnP	Wireless → WPS; System → Remote management; Advanced → UPnP	Off unless specifically needed

If the router belongs to the internet provider

Ask the provider to change the admin password or update firmware if you cannot. If the router no longer receives updates, ask for a replacement.

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TOPIC 4 · SMART CAMERAS AND CCTV

5 minutes · securing cameras · cloud access · placement and consent · retention · buying wisely

What is it?

Cameras protect homes and shops — and, if hijacked, expose them. A secure camera has a changed password, current firmware, MFA on its app and no unnecessary remote access. A **respectful** camera covers only what it needs to, never private spaces of others, and people know it is there. Under the DPDP Act 2023, video of identifiable people is personal data.



Figure 9.6 · Smart cameras and CCTV: secure and respectful

Real-life case · the society gate camera

A housing society installed eight cameras with an online viewing app and left the installer's default password. A resident's teenage son found the live feed listed on a website of exposed cameras. The committee changed all passwords, updated the recorder's firmware, removed public cloud sharing, moved one camera that overlooked a flat's balcony, and put up a notice.

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TOPIC 5 · WEARABLES, SPEAKERS, TVS AND TRACKERS

5 minutes · privacy settings · children's devices · unknown-tracker alerts · stalking

What is it?

Smart speakers listen for a wake word; smart TVs record viewing habits; fitness bands record routes and health data; children's watches share location. Each has privacy settings that are rarely changed. Small Bluetooth trackers, meant for finding keys, can be misused to follow a person; modern phones can alert you to an unknown tracker travelling with you.

Privacy settings on everyday smart devices

Five minutes per device

Device	Check	Why
Smart speaker	Mute button when not needed; review and delete voice history	Recordings can capture private conversations
Smart TV	Turn off viewing-data collection; sign out of shared accounts; app permissions	TVs collect what you watch; logins stay on the TV
Fitness band / smartwatch	Hide location and routes on public profiles; limit sharing	Running routes reveal home and routine
Child tracker / kids' watch	Change default password; only parents in contacts; location to family only	A stranger must never see a child's location
Bluetooth trackers	Turn on “unknown tracker alerts” on your phone	Detects trackers secretly placed in a bag or car

Figure 9.7 · Privacy settings on everyday smart devices

If an unknown tracker is found

Safety first: a tracker placed on someone may be part of stalking or domestic abuse. Note and photograph it and the phone alert; do not go home if the person feels unsafe; contact the police or 181. Disabling it may alert the person who placed it, so plan with the person and the police (Module 7).

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TOPIC 6 · RESPONDING TO SMART-DEVICE INCIDENTS AND OT BASICS

5 minutes · leaked feeds · trackers · botnets · preserve then secure · OT escalation

What is it?

Smart-device incidents combine privacy, safety and evidence. **Preserve first** (screenshots of the app, the leaked page address, the tracker alert), **then isolate and secure** — and check every other device on the same network, because one weak device usually means others.

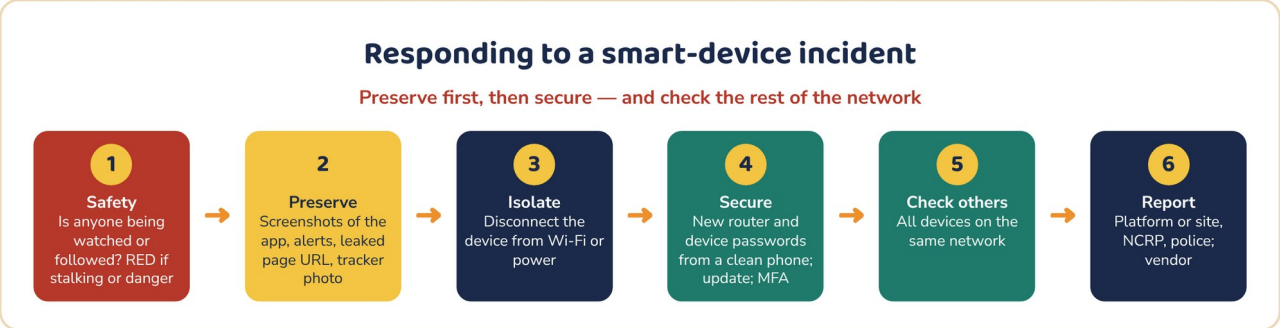


Figure 9.8 · Responding to a smart-device incident



Figure 9.9 · OT in the community

KEY TAKEAWAYS FROM MODULE 9

Takeaway	In one line
Change every default password	On day one, on every device
Router first	Admin password, WPA2/WPA3, firmware, guest network, WPS off
Respectful cameras	Secure, limited view, notice, short retention
Ask “who else can see this?”	For every speaker, TV, watch and tracker
Preserve, isolate, secure, check all	Then report
OT: escalate, never operate	Owner, vendor, IT lead, CERT-In route

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